



PARUL UNIVERSITY
PARUL INSTITUTE OF ENGINEERING AND TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

Weekly Progress Report Card of B. Tech Project (AY:2026-27)

BRANCH: COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)		SEMESTER: 7th	UG: B. Tech
TITLE OF THE DISSERTATION: SecureMail: A Tri-Modal Hybrid Email Security Gateway and Forensics Framework for Zero-Trust Phishing and Optical Quishing Mitigation			
NAME OF DISSERTATION SUPERVISOR - 1: Prof. Akash Patel			
NAME OF DISSERTATION SUPERVISOR - 2 : -			
NAME AND CONTCT NO. OF INTER/MULTI/TRANS SUPERVISOR - 2 (IF ANY):-			
INDUSTRY PARTICULARS(IF ANY):-			
ENROLLMENT NO	NAME OF THE STUDENT	MOBILE NO	
2303031260196	Samir Khurshid	+91 98XXXXXXXX	
2303031260149	M Harshith Chowdary	+91 98XXXXXXXX	
2303031260009	Aman Kumar	+91 98XXXXXXXX	
2303031260145	M. Prem Sai	+91 98XXXXXXXX	

WEEK NO	Meeting Date	DETAILS OF WORK DONE	SIGNATURE OF STUDENT	SIGNATURE OF SUPERVISOR
WEEK 01	08/06/2026	Finalised project title "SecureMail: Tri-Modal Email Security Gateway & Forensics Framework"; formulated problem statement, zero-trust scope and objectives with supervisor.		
WEEK 02	15/06/2026	Conducted literature survey on Secure Email Gateways (SEGs), BEC techniques, homograph attacks, and optical quishing vulnerabilities in modern email clients.		
WEEK 03	22/06/2026	Identified research gap in client-side quishing & DNS authentication bypasses; curated benchmark datasets (Nazario Phishing, Enron, synthetic phishing corpus).		
WEEK 04	29/06/2026	Formulated Software Requirements Specification (SRS); finalized modular decoupled architecture (FastAPI backend, Chrome Extension, Web SOC dashboard).		
WEEK 05	06/07/2026	Developed core asynchronous email ingestion & MIME parsing engine (RFC 822/5322); implemented header extraction, body normalization, and URL/attachment parsers.		

WEEK 06	13/07/2026	Implemented Modal 1: Real-Time DNS-over-HTTPS (DoH) engine for cryptographic authentication (SPF record validation, DKIM signature verification, DMARC alignment).		
WEEK 07	20/07/2026	Implemented Modal 2: IDN Homograph & Typosquatting defense engine; developed Punycode decoder, Levenshtein distance matching, and visual confusable detection.		
WEEK 08	27/07/2026	Implemented Modal 3: Optical Quishing Detonation Pipeline; integrated OpenCV & pyzbar for QR extraction, headless redirect unshortening, and sandbox URI analysis.		
WEEK 09	03/08/2026	Developed hybrid local ML phishing classifier (TF-IDF vectorizer + ensemble model) and engineered weighted multi-factor heuristic risk-scoring algorithm.		
WEEK 10	10/08/2026	Integrated external threat intelligence APIs (VirusTotal, AbuseIPDB); implemented GeoIP resolver and SMTP hop latency trajectory tracker for forensic mapping.		
WEEK 11	24/08/2026	Engineered Chrome Manifest V3 browser extension; built content scripts for Gmail DOM parsing and real-time in-pane threat warning banner injection.		
WEEK 12	31/08/2026	Developed glassmorphic SOC frontend dashboard; implemented live threat metrics, MITRE ATT&CK mapping, anonymous scan quota, and RBAC admin controls.		
WEEK 13	07/09/2026	Executed comprehensive empirical validation on 123 benchmark test cases; compiled metrics (97.87% accuracy, 99.06% precision, 96.72% recall, 0.94% FPR).		
WEEK 14	14/09/2026	Began drafting IEEE 2-column format research paper: authored abstract, introduction, literature survey, and tri-modal threat detection methodology.		
WEEK 15	21/09/2026	Completed research paper experimental results, ablation analysis, and conclusion; compiled IEEE bibliography; drafted comprehensive B.Tech project report.		
WEEK 16	28/09/2026	Finalized research paper formatting; conducted official Turnitin/iThenticate evaluation (1% similarity, AI verified human authorship); prepared dissertation defense.		
Sign. of PG Coordinator Comments from H.O.D: Satisfactory/ Unsatisfactory Sign. of HOD				